PERFECTNESS OF A PSEUDO-COHERENT COMPLEX OVER A FLAT LOCAL EXTENSION WITH REGULAR FIBRE, FROM ONE FIBRE COMPUTATION AT THE CLOSED POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 20228-20244 (canonical tag 09PC) and the counted proof is the proof environment at source lines 20246-20273, which immediately follows it and is its own proof.

Editorial note. The proof refers three times: to example-cohomology-complex-tensored and lemma-check-perfect-pointwise inside this chapter, and to algebra-proposition-regular-finite-gl-dim of another chapter. All three are navigation pointers with their roles recorded in the external reference reasons; the displayed computation of the fibre at the closed point and the degree estimate are inside the counted span.

Editorial note. The sentence at source line 20226, announcing that the lemma can be used to find perfect complexes over regular local rings, is outside the counted spans.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- The derived tensor product with the residue field at the closed point of a local ring detects perfectness of a pseudo-coherent complex, through the vanishing of the cohomology of that fibre outside the tor amplitude range. invoked at proof line 20272, where the vanishing obtained for j outside [a - d, b] is turned into the conclusion. Converts the single fibre computation at the closed point into perfectness over the local ring B.
- A regular local ring has finite global dimension equal to its dimension, so the fibre B over m_A B has global dimension d. invoked at proof lines 20267-20269 by reference to the algebraic result that regular rings have finite global dimension. Supplies the bound d that shifts the tor amplitude range and makes the spectral sequence argument terminate.

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- For a bounded above complex of flat modules the ordinary tensor product computes the derived tensor product, and the spectral sequence of the example on cohomology of a tensor product of complexes relates the cohomology of the fibre to that of K tensor_A kappa(m_A). used at proof lines 20257-20261 and 20265-20267, where the three equalities of the display are justified and the spectral sequence is applied. Produces the identification K tensor^L_B kappa(m_B) = (K tensor_A kappa(m_A)) tensor^L over B over m_A B kappa(m_B) and hence the degree estimate.

Source reference. Original remark 061Z: sources/stacks-more-algebra/pinned-original-source.tex, source lines 15159–15159. The reference names the spectral sequence of a tensor product of complexes used at proof lines 20265-20267 to pass from the cohomology bound on the fibre over A to the bound on the fibre over B; the identification of the fibre and the degree bookkeeping around it are inside the counted span.

Source reference. Original lemma 068V: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20032–20032. The reference is the criterion invoked at proof line 20272 to conclude perfectness from the vanishing of the cohomology of the fibre; the vanishing itself is proved inside the counted span.

Source reference. Original proposition 00O7: sources/stacks-chow/tags.txt, source lines 885–885. The reference supplies the standard fact that a regular ring has finite global dimension, used at proof lines 20267-20269 to bound the global dimension of the regular fibre by its dimension d; the spectral sequence argument that consumes the bound is inside the counted span.

Lemma 0.2. *Let $A \to B$ be a local ring homomorphism. Let $a, b \in \mathbf{Z}$. Let $d \geq 0$. Let $K^\bullet$ be a complex of B-modules. Assume*

1. *the ring map $A \to B$ is flat,*
2. *the ring $B/\mathfrak{m}_A B$ is regular of dimension d,*
3. *$K^\bullet$ is pseudo-coherent as a complex of B-modules, and*
4. *$K^\bullet$ has tor amplitude in $[a, b]$ as a complex of A-modules, in fact it suffices if $H^i(K^\bullet \otimes^{\mathbf{L}}_A \kappa(\mathfrak{m}_A))$ is nonzero only for $i \in [a, b]$.*

Then $K^\bullet$ is perfect as a complex of B-modules with tor amplitude in $[a - d, b]$.

Proof. By (3) we may assume that $K^\bullet$ is a bounded above complex of finite free B-modules. We compute

$$\begin{aligned} K^\bullet \otimes^{\mathbf{L}}_B \kappa(\mathfrak{m}_B) &= K^\bullet \otimes_B \kappa(\mathfrak{m}_B) \\ &= (K^\bullet \otimes_A \kappa(\mathfrak{m}_A)) \otimes_{B/\mathfrak{m}_A B} \kappa(\mathfrak{m}_B) \\ &= (K^\bullet \otimes_A \kappa(\mathfrak{m}_A)) \otimes^{\mathbf{L}}_{B/\mathfrak{m}_A B} \kappa(\mathfrak{m}_B) \end{aligned}$$

The first equality because $K^\bullet$ is a bounded above complex of flat B-modules. The second equality follows from basic properties of the tensor product. The third equality holds because $K^\bullet \otimes_A \kappa(\mathfrak{m}_A) = K^\bullet/\mathfrak{m}_A K^\bullet$ is a bounded above complex of flat $B/\mathfrak{m}_A B$-modules. Since $K^\bullet$ is a bounded above complex of flat A-modules by (1), the cohomology modules H^i of the complex $K^\bullet \otimes_A \kappa(\mathfrak{m}_A)$ are nonzero only for $i \in [a, b]$ by assumption (4). Thus the spectral sequence of Example 061Z and the fact that $B/\mathfrak{m}_A B$ has finite global dimension d (by (2) and Algebra, Proposition 00O7) shows that $H^j(K^\bullet \otimes^{\mathbf{L}}_B \kappa(\mathfrak{m}_B))$ is zero for $j \notin [a - d, b]$. This finishes the proof by Lemma 068V. □